Test-retest reproducibility of elbow goniometric measurements in a rigid double-blinded protocol: Intervals for distinguishing between measurement error and clinical change

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Increasingly, goniometry of elbow motion is used for qualification of research results. Expression of reliability is in parameters not suitable for comparison of results. We modified Bland and Altman's method, resulting in the smallest detectable differences (SDDs). Two raters measured elbow excursions in 42 individuals (144 ratings per test person) with an electronic digital inclinometer in a classical test-retest crossover study design. The SDDs were 0 ± 4.2° for active extension; 0 ± 8.2° for active flexion, both without upper arm fixation; 0 ± 6.3° for active extension; 0 ± 5.7° for active flexion; 0 ± 7.4° for passive flexion with upper arm fixation; 0 ± 10.1° for active flexion with upper arm retroflexion; and 0 ± 8.5° and 0 ± 10.8° for active and passive range of motion. Differences smaller than these SDDs found in clinical or research settings are attributable to measurement error and do not indicate improvement. (J Shoulder Elbow Surg 2007;16:788-794.)

A major proportion of daily practice of orthopaedic surgeons concerns fracture treatment and mobility problems of joints. Valid evaluation of therapeutic success is paramount. Established measures for treatment outcome are duration of fracture healing, number of delayed unions, pseudarthrosis and infections, pain scores, weight-bearing capacity, and range of motion (ROM) of the treated and adjacent joints. Objective assessment of joint motion is essential for identification of treatment failure and the need for therapeutic intervention.

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The gold standard for evaluation of joint motion is measurement of maximum joint excursion by radiography,[6,7,23] which has the disadvantage of being time-consuming and exposes patients to radiation. In clinical practice, the most used noninvasive alternative to x-ray measurement of joint angles is goniometry. Both traditional parallelogram goniometers and sophisticated forms like gravity goniometers are widely used.[14]

In research settings, digital gravity goniometry is increasingly applied for expression of success of treatment, and good levels of reproducibility have been claimed.[2,21,22] However, the statistical methods used to express intra- and inter-rater reproducibility in these studies (Pearson's correlation coefficient, intraclass correlation coefficient, or the coefficient of variation) are merely statistical tools to express correlation but lack clinical applicability. In the present study, we studied the intra- and inter-rater reproducibility of gravity goniometry in elbow measurements, using a rigid double-blinded protocol. Moreover, we introduce a statistical approach that expresses reproducibility in a clinically applicable way, defining reproducibility for digital goniometry in confidence limits. These confidence limits can be used for reference in future research settings. Differences found with digital goniometry smaller than these intervals should be interpreted with caution, as these are most likely the result of measurement error.

MATERIALS AND METHODS

Subjects

The sample size calculated for a 90% power of testing and a 1% level of significance with a true correlation of .60 ($\alpha = .01$, $\beta = .10$) was a minimum of 30 subjects.[11] In the present study, 42 healthy volunteers were recruited (23 males; mean age, 22.1 years; range, 19-27 years). Participation followed informed consent. Criteria for exclusion were previous or current complaints of the elbow or shoulder symptoms and previous injuries or prior operations on the elbow or shoulder. Left- and right-handed persons were measured on the dominant side.[15]

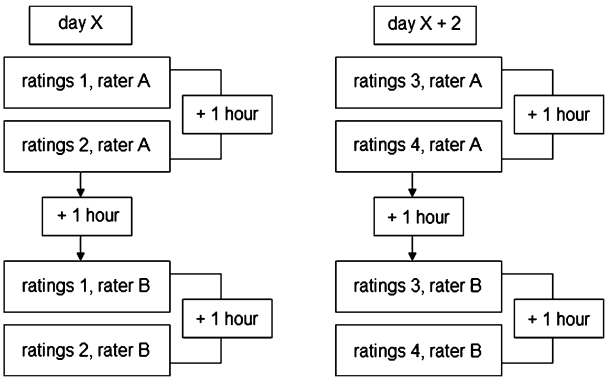


Figure 1 Time-rating sessions relation. One hour rest between each session, two days rest between session 2 and 3, starting rater (A or B) randomized.

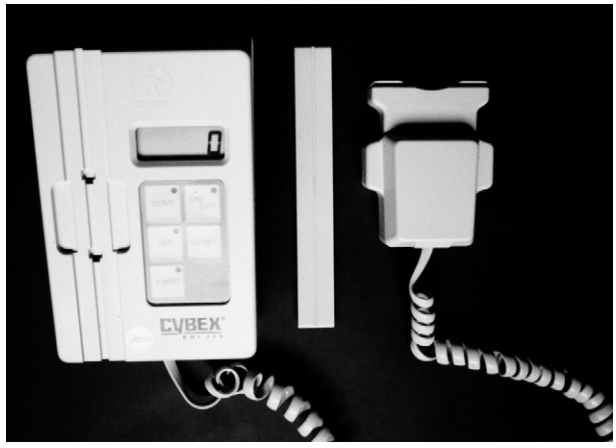


Figure 2 Electronic Digital Inclinometer (EDI 320).

Protocol

We used the classical test-retest crossover design (Figure 1) that was executed by 2 raters (A and B), who were experienced users and meticulously trained with the device and the protocol. Both raters and test subjects were blinded to the results of measurements. Subjects were studied during 4 sessions for each rater, with 2 sessions being executed on the first measurement day and 2 sessions after 2 days at the same hour of the day (Figure 1). During each session, 3 measurements were performed for each joint excursion, resulting in a total of 144 measurements per test subject. Subjects were studied on the same day by both raters (1-hour rest in between). The rater performing the 1st measurements session remained constant for each subject but was equally distributed between subjects, after randomization by sealed envelope drawing (rater A was first in 22 subjects and rater B in 20).

Materials

Joint excursion was measured using the electronic digital inclinometer (EDI) type EDI 320 (Cybex Inc, Ronkonkoma, NY). This device is gravity dependent and indicates motion on a 360° scale. The EDI 320 consists of a hand-held unit and a portable display unit with an integral rechargeable power source (Figure 2). The EDI 320 has 3 measurement sequences: continuous mode for continuous registration, single reference mode for recording gross motion, and compound reference mode that allows determination of separate angles within a greater range of motion (ie, precise joint angle detection). In the present study, all measurements were performed in the single reference mode for joint position measurements.

Setting and standardization

The rating and data collection sessions were held at a university kinematics science laboratory. A practice session was held to establish a standardized testing procedure. Measurements were performed at right angles to the gravity field at all times. In order to standardize testing procedures, anatomic reference points were marked with a skin marker with the elbow flexed at 90°. Reference points were 3 cm distal to the olecranon on the dorsal surface of the ulna and 3 cm proximally at the imaginary line in between epicondyles.[19] Before the measurements of joint excursion were performed, full joint motion was executed 3 times to diminish stretching and learning biases.[3]

Measurements

Elbow ROM is influenced by movements of the shoulder joint.[18,20,24] Fixation of the upper arm should hypothetically result in diminished shoulder movements and, therefore, improve reproducibility. To evaluate the need for fixation of the upper arm, 3 methods for testing active maximum flexion were evaluated. The first method (A) consisted of a setting with the upper arm fixed to a fixation block and the test persons in sitting position with the shoulder in a horizontal plane to the fixation block (20 x 22 x 15 cm), resulting in 90° forward flexion of the shoulder. The EDI was placed on the dorsal surface of the distal upper arm (Figure 3). The second method (B) was with the test persons standing straight. The upper arm was placed adducted against the body with the shoulder in 0° flexion, and with the hand and forearm in midpronation. The elbow was flexed from this position while keeping the upper arm in the same position (Figure 4). An alternate method (C) of measurement (theoretically resulting in larger maximum flexion by stretching the biceps brachii) was performed with the shoulder in retrofexion, without abduction and rotation.[18,20] Again, subjects were supine and flexion was accomplished with the upper arm held still (Figure 5). Measurement of elbow active maximum flexion and passive maximum flexion was performed from a starting position of active maximum extension and vice versa, as this has been shown to result in the greatest joint excursion. Passive maximum flexion was only determined by method A because it is impossible to keep the upper arm and shoulder in the same position while applying force to the forearm.

Statistics

Reproducibility can be expressed by a variety of methods, of which the Pearson product-moment correlation coefficient (Pearson's r) and the intraclass correlation



Figure 3 Active maximum flexion method A.

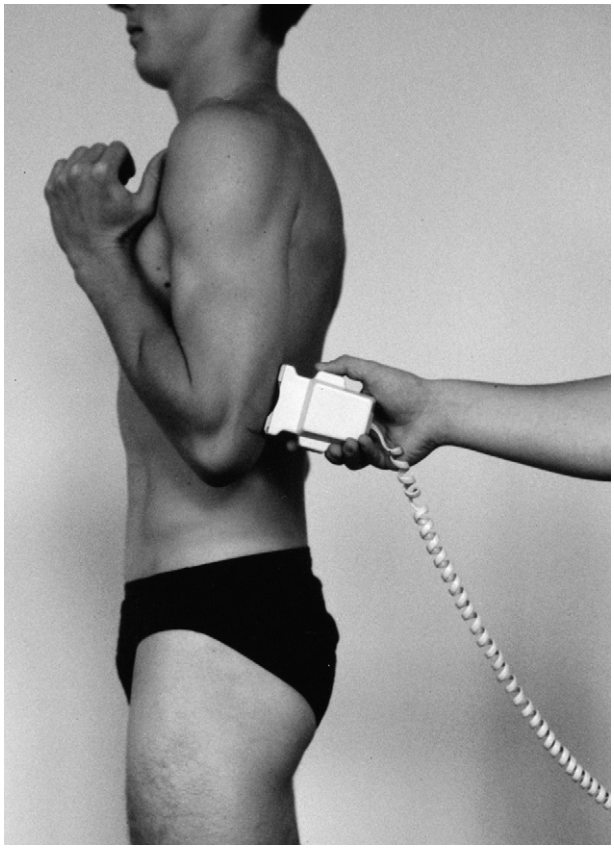


Figure 4 Active maximum flexion method B.



Figure 5 Active maximum flexion method C.

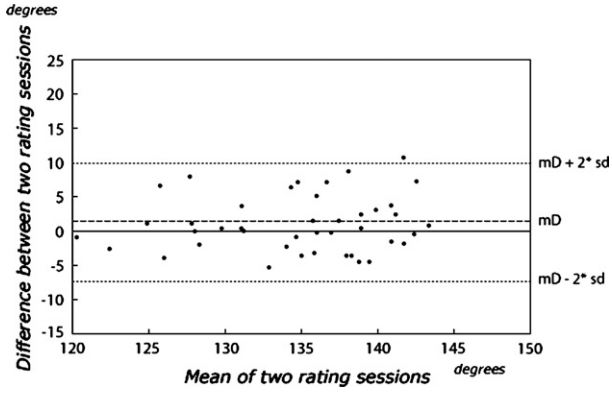


Figure 6 Example of a Bland-Altman scatterplot, intra- rater B, extreme sessions, active maximum flexion method C. mD: mean difference. sd: standard deviation. Area within mD ± 2 * sd represent SDD for given measurement method.

coefficients (ICCs) are the most commonly used. Pearson's *r* considers random, but not systematic, error between test scores.[9,10] High reproducibility depends on small random error and absence of systematic error between repeated measurements.[17] The ICC takes both random and systematic errors into account and is, therefore, a better measure than Pearson's correlation.[9] However, both measures strongly depend not only on the amount of systematic and random error, but also on the heterogeneity of the study population,[8,13] which makes it difficult to compare different reliability studies and use these coefficients for clinical decision making. We, therefore, prefer to use the

Table I Clinically most informative inter- and intra-rater SDDs based on extreme sessions

	Inter-rater SDD, 1m	Pearson's	Inter-rater SDD, m3	Pearson's	Intra-rater SDD, m3	Pearson's
	0 ± (|mD| + 1.96*sd)	r, 1m	0 ± (|mD| + 1.96*sd)	r, m3	0 ± (|mD| + 1.96*sd)	r, m3
AME no fixation	0 ± 7.6	0.82	0 ± 4.2	0.89	0 ± 5.7	0.89
AME with fixation	0 ± 8.5	0.75	0 ± 6.3	0.84	0 ± 5.7	0.87
AMF method A	0 ± 7.1	0.77	0 ± 5.7	0.85	0 ± 5.4	0.87
AMF method B	0 ± 10.5	0.66	0 ± 8.2	0.77	0 ± 7.8	0.82
AMF method C	0 ± 13.7*	0.67	0 ± 10.1*	0.79	0 ± 7.2	0.85
PMF with fixation	0 ± 8.5	0.71	0 ± 7.4	0.78	0 ± 6.1	0.85
AROM method A	0 ± 11.3	0.78	0 ± 8.5	0.87	0 ± 8.1	0.89
AROM method B	0 ± 13.1	0.75	0 ± 10.2	0.84	0 ± 9.0	0.88
AROM method C	0 ± 16.7*	0.75	0 ± 12.5*	0.89	0 ± 9.1	0.88
PROM with fixation	0 ± 14.1*	0.76	0 ± 10.8*	0.84	0 ± 7.9	0.89

AMF, active maximum flexion; AME, active maximum extension; PMF, passive maximum flexion; ROM, range of motion.
*Marked intervals are corrected for mean difference (mD); other intervals are uncorrected. Intra-rater intervals as a mean of the SDD per rater, under assumption of equality of raters.

limits of agreement as proposed by Bland[5] and Altman,[1] which is based on the mean and standard devation (SD) of the difference between two ratings of the same patient and, thereby, has 3 advantages. First, it does not systematically depend on the heterogeneity of the study population. Second, it quantifies the amount of measurement error in units of the measurement scale used, for example, in degrees. Third, it is only minimally influenced by the number of raters for determination of inter-rater reliability.

A measure of agreement must indicate the possible size of the difference between 2 raters (or ratings by the same rater) due to systematic as well as random measurement errors. The mean difference between 2 raters, if statistically significant, indicates systematic error in either rater. The SD of the difference between raters indicates the amount of random error, since the variance of this difference is equal to the sum of the measurement error variance of rater A and that of rater B. Bland and Altman proposed to calculate the 95% limits of agreement (mean of the difference ± 1.96* SD), which indicates the maximum difference between raters that can be expected due to systematic and random measurement error.[1,4,5] This Bland and Altman method can be visualized by plotting the difference between the first and second ratings against the mean of the first and second ratings (Figure 6). Apart from sampling error, the mean of the *N* individual differences between the 2 raters must be zero, unless there is a systematic difference between both raters. This can be tested with Student's paired *t* test. If the mean difference is significant, then the confidence limits of agreement must take this into account, as already done by the Bland and Altman method. However, if these limits are to be applied in clinical practice with other raters than the 2 under study, then the sign of the mean difference (mD $>$ 0 or $<$ 0) is not known beforehand.

Taking the mD of our raters as best guess of the mean difference that may be expected to occur in clinical practice with other raters of similar experience and training, and taking into account that the sign of the mD ($>$0 or $<$0) for 2 such other raters is not known beforehand, we obtain the following 95% confidence limits for individual differences between 2 raters:-|mD| - 1.96*SD to +|mD| + 1.96*SD.

These corrected confidence limits can also be termed the smallest detectable difference (SDD), as they represent the (95%) threshold for change that can be detected by the EDI 320 beyond systematic and random measurement error.[4,9] These intervals assume that, apart from a possible systematic difference reflected by a mD unequal to zero, the raters are equally reliable. Consequently, the intrarater correlation between 2 measurements should be about the same for both raters. This equality of intrarater correlations and SDs between raters was confirmed by the data (see, for example, the Appendix, where raters A and B have very similar SDs and correlations).

Reproducibility was determined on the basis of (a) the mean of all 3 measurements per patient per session (in the sequel denoted as m3) and (b) the first measurement per patient per session (denoted 1 m). Note that m3 will give higher a higher reproducibility than 1 m due to the Spearman-Brown effect, that is well known in psychometric literature.[8] Using each of these 2 bases, the following intra- and inter-rater comparisons were made to evaluate reproducibility.

Intrarater. Same day (for each day per rater: mD, SD, SDD and *r* between both sessions for this rater on this day were computed, followed by taking the average of each statistic over days) (Table 1 and Appendix). Different day sessions (ie, measurement session 1 vs 3, 1 vs 4, 2 vs 3, 2 vs 4), source of disagreement: random errors. Extreme sessions (using only the first and the last measurement session, ie,1 and 4), sources of disagreement: random errors plus time effects that may occur between repeated visits of a patient to the same clinician.

Inter-rater. Same sessions (for each of the 4 sessions per rater, mD, sd, SDD, and r between raters were computed, and each of these statistics was then averaged over the 4 sessions, ie, session 1 rater A vs 1 rater B, 2 vs 2, 3 vs 3, 4 vs 4), sources of disagreement: random errors plus systematic difference between raters. First sessions, sources of disagreement: as for same sessions, but first sessions may be clinically more realistic (ie, using only: session 1 rater A vs 1 rater B). Extreme sessions (ie, using only: session 1 rater A vs 4 rater B, session 4 rater A vs 1 rater B), sources of disagreement: random errors plus systematic difference

between raters plus time effects between repeated visits of a patient.

The data were analyzed using SPSS version 12.0 (Statistics Package for the Social Sciences, SPSS Inc, Chicago, Il). For each outcome, the presence of a systematic mean difference between raters, or between sessions within a rater, was tested with the Student *t* test for paired samples, using a 2-tailed α of 5%. Although the risk of a type I error (ie, a false significance) is increased by multiple testing, we did not choose a smaller α to prevent type II errors (ie, false nonsignificance), because a type II error would lead to a SDD that is too small.

Additionally, we tested the mean outcome difference between male and female patients with the Student *t* test for unpaired samples, using as outcome the average of the 24 (8 sessions, * 3 measurements) replications per person.

Note that the presence of such a difference has no implication for the calculation of the SDDs, as SDDs are based on intrapatient differences between replications and, thereby, unaffected by the amount of variation between patients. In contrast, measures such as the ICC and Pearson's correlation become higher as variation between patients increases.[8]

RESULTS

The test-retest protocol was executed as planned, and all subjects were measured accordingly. A systematic inter-rater difference (ie, mD significantly different from 0) was found for elbow active maximum flexion method C, the AROM determined using method C and the PROM. Therefore, these SDDs were corrected (see Materials and Methods section). Further scrutiny of these confidence limits showed that this systematic difference between raters was $\leq 2.1^\circ$. For all other intra- and inter-rater mD comparisons in all other joint position replications, Student's *t* test for paired samples did not result in a significant mean difference.

Table I shows the SDDs of the clinically most informative method of comparison of inter-rater agreement, that is, extreme sessions. Systematic differences in mD resulted in correction of the intervals with -|mD| or +|mD| as marked. It could be determined that SDDs narrow by 1.1° to 4.2° when the mean of 3 measurements was taken compared to 1 single measurement. Measurement of active maximum flexion of the elbow conducted with the upper arm on a fixation block (method A) resulted in a more narrow SDD ($0 \pm 5.7^\circ$), reflecting better reproducibility than with methods B ($0 \pm 8.2^\circ$) and C ($0 \pm 10.1^\circ$). For active maximum extension, the SDD was slightly greater with the fixation block ($0 \pm 4.2^\circ$ without fixation (method B) vs $0 \pm 6.3^\circ$ with fixation [method A]). Calculation of AROM out of active maximum flexion and extension enlarged SDDs by 2 - 2.8° when compared to single joint excursion SDDs.

Table I also shows the level of inter- (extreme sessions) and intra-rater (extreme sessions) reproducibility estimated using Pearson's *r*. Pearson's *r* values between 1.0 and .90 are generally accepted as excellent, .89-.80 as good, .79-.70 as fair, and lower than .70 as poor.[16] Nearly all correlations measured in the present study between means of measurement per session (m3) were rated with excellent or good reproducibility (data not shown but available in the Appendix; see www.jshoulderelbow.org). Only active maximum flexion in methods B and C resulted in poor correlations (.60 -.70) based on the first measurement (1 m). The level of reproducibility between measurements based on only the first measurement was, for most joint positionings, fair to good. Pearson's correlation based on the m3 method for inter-rater comparisons was up to .14 higher than Pearson's *r* based on the 1 m method, depending on the outcome at hand.

Of further interest is that our results showed that active maximum flexion was larger for females than for males (using as outcome the mean of all 24 replications per patient, $P < .05$, two-tailed unpaired *t* test; Table II). For the total study population elbow passive maximum flexion was significantly larger than in active maximum flexion (using as outcome the mean of all 24 replications per patient, $P < .001$, two-tailed paired *t* test; Table II).

DISCUSSION

The main aim of our study was to find clinical application for reproducibility in goniometric readings with an EDI 320 gravity goniometer in a simple protocol. The present data show that good standardization of a goniometric measurement protocol results in highly comparable absolute intra-rater SDDs for both raters. These are also explained by the fact that we used healthy young individuals to exclude variability, and we minimized learning and stretching bias by a premeasurement practice session. No statistical significant difference in SDDs of active maximum extension was found for methods of measurement with or without fixation of upper arm. However, we did find a difference in active maximum extension of 2.7° between methods, with nonfixation leading to further joint excursion (Table II). This may be explained by placement of the EDI 320 goniometer on the dorsal side with muscle relaxation in method A opposed to method B, where there is no full relaxation of the triceps.

Method B resulted in higher reproducibility than method C for measurements of active maximum flexion (SDD $0 \pm 8.2^\circ$ vs $0 \pm 10.1^\circ$). Method B also rendered further active maximum flexion than method C (137.2° vs 135.5°), although clearly not statistically significant. This rejects the theory that elbow flexion with the shoulder in retroflexion would result in greater excursion due to the stretch on the biceps.

Fixation of the upper arm (method A) narrowed confidence limits with 2.5° and 4.4° as opposed

Table II Differences between sexes in elbow motion

	Group	Mean	SD	Min	Max
AME no fixation	all	−3.5	5.9	−12.7	11.3
	♂	−3.6	5.9	−12.7	6.5
	♀	−3.2	6.1	−10.6	11.3
AME with fixation	all	−0.8	5.1	−8.5	16.0
	♂	−1.4	4.6	−8.5	8.0
	♀	−0.1	5.6	−7.4	16.0
AMF method A	all	141.3**	4.9	131.0	150.7
	♂	139.9*	5.1	131.0	150.7
	♀	142.9*	4.2	134.3	149.5
AMF method B	all	137.2	5.6	125.4	147.5
	♂	135.2*	5.7	125.4	144.9
	♀	139.6*	4.4	133.4	147.5
AMF method C	all	135.5	6.0	119.9	147.2
	♂	132.7*	5.8	119.9	143.7
	♀	138.9*	4.4	131.1	147.2
PMF	all	146.0**	5.0	132.9	156.5
	♂	145.0	5.6	132.9	156.5
	♀	147.1	3.9	139.5	155.0
AROM method A	all	142.0	7.9	118.2	156.5
	♂	141.3	7.6	124.2	150.1
	♀	142.9	8.3	118.2	156.5
AROM method B	all	140.6	8.6	119.3	156.9
	♂	138.8	8.5	119.3	149.8
	♀	142.8	8.3	124.0	156.9
AROM method C	all	138.9	8.9	117.5	151.5
	♂	136.3*	9.0	117.5	148.6
	♀	142.1*	7.9	119.8	151.5
PROM	all	146.8	7.8	125.8	159.8
	♂	146.4	7.9	129.6	155.4
	♀	147.2	7.9	125.8	159.8

Using as outcome the average of all 24 replications per patient, with N = 42 patients in total.

AMF, active maximum flexion; AME, active maximum extension; PMF, passive maximum flexion; ROM, range of motion.

*$p < 0.05$ between sexes.

**$p < 0.001$ between AMFP method A and PMFP.

to methods B and C for active maximum flexion. Our results suggest that elbow joint motion should be measured using fixation of the upper arm. It can be hypothesized that this method would even improve reproducibility more when used in the elderly, but this needs further study. A small difference in reproducibility was found for active maximum flexion versus passive maximum flexion SDDs (confidence limits $0 \pm 5.7°$ vs $0 \pm 7.4°$). As expected, maximum joint excursion was larger for passive maximum flexion (156.5° vs 150.7°).[12]

Literature indicates that ROM of the elbow joint for females is larger than for men. This difference has always been attributed to a larger excursion for extension. With calculation of ROM in our method, we also found a larger ROM for females (142.1° vs 136.3°) but based on a larger excursion for active maximum flexion in method C (138.9° vs132.7°) (Table II). Determination of ROM out of active maximum flexion and extension resulted, as expected, in enhancement in spread of confidence limits with a range of 2° - 2.8°.

Calculation of reproducibility using the mean of 3 measurements resulted in narrowing of the SDD with 1.1° - 4.2°, consequently, also improving reproducibility. Using only 1 measurement for clinical practice reduces reproducibility and, with it, the power to detect true alterations in capacity for joint motion in patients, due to the Spearman-Brown effect .[8]

Calculation of the SDD using Bland-Altman's method, as used in this study, is useful for clinical practice with individual patients. In contrast to reproducibility measures like Pearson's *r* or ICC, the SDD can be used in clinical practice, as it expresses the threshold for change that can be detected by the EDI 320 beyond measurement error. SDD is, therefore, a clinically applicable statistic that gives the physician a strong tool that can be used to determine if the change in joint excursion of an individual patient is based on random error of the measurement or based on therapeutic success or failure. It can be hypothesized that determination of SDDs in clinical practice will result in broader confidence limits, as ours were the result of strict standardization in young, healthy volunteers without joint complaints.

As expected, we found better intra- than inter-rater reproducibility scores. Not surprisingly, our data show that the ideal evaluation of a change in joint motion is by measurements performed by the same rater at a fixed time using a standardized protocol, performing at least 3 measurements. However, in clinical practice, revision is not always by the same rater.

As mentioned, goniometry is increasingly used for qualification of therapeutic success in literature. For future research studies with use of digital goniometry, conclusions about improvement, even if the result of statistical testing between two intervention groups, should be made with caution. A statistical difference for alteration in joint motion in degrees smaller than the SDDs presented can technically not be measured with digital goniometry, as it is within the smallest detectable difference for the method of measurement. We advise use of the SDDs in Table I for judgements about clinical change. Differences smaller than these SDDs are probably attributable to measurement error and do not indicate improvement.

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Appendix Intra- and inter-rater comparisons

	1st measurement (1m)			Mean of 3 measurements (m3)		
	mD	**1.96 * sd**	**r**	**mD**	**1.96 * sd**	**r**
AME no fixation						
Inter-rater						
Same sessions	−1.0	7.6	.82	0.1	5.6	.90
First sessions	−0.5	8.0	.81	0.5	6.2	.88
Extreme sessions	−0.1	7.6	.82	−0.3	4.2	.89
Intra-rater A						
Same day sessions	−0.1	7.7	.81	−0.3	4.3	.94
Extreme sessions	0.3	7.5	.82	−0.3	5.4	.90
Intra-rater B						
Same day sessions	0.1	7.3	.83	0.0	5.6	.90
Extreme sessions	−1.0	6.3	.88	−0.7	6.0	.89
AME with fixation						
Inter-rater						
Same sessions	−0.7	8.0	.75	−0.3	5.9	.85
First sessions	−0.7	9.3	.69	−0.4	6.5	.84
Extreme sessions	−0.8	8.5	.75	−0.4	6.3	.84
Intra-rater A						
Same day sessions	−0.4	7.0	.82	−0.4	5.2	.89
Extreme sessions	−0.1	6.9	.83	−0.3	5.0	.90
Intra-rater B						
Same day sessions	−0.1	7.6	.79	−0.2	5.8	.86
Extreme sessions	−0.3	8.8	.74	−0.3	6.4	.84
AMF method A						
Inter-rater						
Same sessions	0.8	6.7	.81	0.6	5.6	.86
First sessions	0.7	6.0	.84	0.6	5.6	.86
Extreme sessions	0.6	7.1	.77	0.6	5.7	.85
Intra- rater A						
Same day sessions	−0.1	5.8	.84	−0.1	5.1	.87
Extreme sessions	0.2	6.2	.82	0.3	5.4	.86
Intra- rater B						
Same day sessions	−0.1	6.3	.84	0.0	4.3	.93
Extreme sessions	−0.1	6.7	.82	0.3	5.4	.88
AMF method B						
Inter-rater						
Same sessions	0.4	10.1	.68	0.7	7.6	.81
First sessions	−0.7	11.9	.56	−0.2	7.7	.80
Extreme sessions	0.3	10.5	.66	0.6	8.2	.77
Intra-rater A						
Same day sessions	−0.4	10.3	.69	−0.4	6.1	.88
Extreme sessions	−1.3	8.7	.77	−1.2	7.7	.81
Intra-rater B						
Same day sessions	1.3	8.4	.76	0.8	6.1	.86
Extreme sessions	0.6	9.4	.72	0.4	6.9	.82
AMF method C						
Inter-rater						
Same sessions	1.5	11.7	.68	1.6	7.8	.82
First sessions	0.7	12.4	.66	1.0	8.2	.80
Extreme sessions	1.9	11.8	.67	1.6	8.5	.79
Intra-rater A						
Same day sessions	−0.5	8.9	.80	−0.3	6.6	.87
Extreme sessions	−1.2	8.9	.81	−0.5	6.4	.89
Intra-rater B						
Same day sessions	1.3	11.8	.69	0.9	7.1	.84
Extreme sessions	1.2	14.4	.53	0.8	8.0	.80

(*Continued*)

Appendix Intra- and inter-rater comparisons (*Continued*)

	1st measurement (1m)			Mean of 3 measurements (m3)		
	mD	**1.96 * sd**	**r**	**mD**	**1.96 * sd**	**r**
PMF						
Inter-rater						
Same sessions	1.1	8.4	.72	0.8	6.0	.84
First sessions	0.6	7.7	.74	0.3	5.6	.88
Extreme sessions	1.1	8.5	.71	0.8	7.4	.78
Intra-rater A						
Same day sessions	−0.1	6.8	.81	−0.2	4.5	.91
Extreme sessions	−0.3	6.9	.78	0.1	5.9	.85
Intra-rater B						
Same day sessions	0.0	7.2	.80	0.1	5.6	.87
Extreme sessions	0.7	8.2	.76	1.0	6.3	.85
AROM method A						
Inter-rater						
Same sessions	1.4	10.3	.75	1.0	8.0	.89
First sessions	1.4	11.8	.76	0.9	9.2	.85
Extreme sessions	1.3	11.3	.78	0.9	8.5	.87
Intra-rater A						
Same day sessions	0.1	9.1	.86	0.2	7.7	.89
Extreme sessions	0.3	9.0	.86	0.6	7.8	.89
Intra-rater B						
Same day sessions	−0.1	9.4	.86	0.2	7.5	.91
Extreme sessions	0.1	11.2	.80	0.6	8.3	.88
AROM method B						
Inter-rater						
Same sessions	0.7	12.5	.78	0.9	9.4	.87
First sessions	−1.2	13.6	.73	−0.7	9.7	.85
Extreme sessions	0.4	13.1	.75	0.3	10.2	.84
Intra-rater A						
Same day sessions	−0.2	11.7	.81	−0.1	6.8	.93
Extreme sessions	−1.6	10.7	.84	−0.7	9.3	.87
Intra- rater B						
Same day sessions	1.0	11.1	.81	0.8	8.6	.88
Extreme sessions	1.5	10.7	.83	1.3	8.7	.88
AROM method C						
Inter-rater						
Same sessions	1.8	13.5	.78	1.8	9.3	.88
First sessions	0.2	13.9	.78	0.6	9.9	.85
Extreme sessions	2.0	14.7	.75	1.4	11.1	.83
Intra-rater A						
Same day sessions	−0.4	11.7	.82	0.0	7.9	.91
extreme sessions	−1.5	12.0	.82	−0.1	8.4	.91
Intra-rater B						
Same day sessions	1.1	14.3	.76	0.9	8.4	.90
extreme sessions	2.1	15.2	.74	1.6	9.7	.87
PROM method A						
Inter-rater						
Same sessions	1.8	11.8	.77	1.3	8.4	.87
First sessions	1.3	11.7	.77	0.7	7.9	.90
Extreme sessions	1.9	12.2	.76	1.2	9.6	.84
Intra-rater A						
Same day sessions	0.2	10.3	.81	−0.2	6.6	.91
Extreme sessions	−0.2	11.6	.74	0.4	8.1	.87
Intra-rater B						
Same day sessions	−0.1	9.9	.85	0.2	7.4	.91
Extreme sessions	1.0	11.0	.82	1.3	7.7	.91

AMF, active maximum flexion; AME, active maximum extension; PMF, passive maximum flexion; ROM, range of motion.